

```
#include <stdio.h>

#define MAX 20

int n_states, n_symbols;
int transition[MAX][MAX];
int final_states[MAX], n_final;
int removed[MAX] = {0};

int is_final(int s) {
    for (int i = 0; i < n_final; i++) {
        if (final_states[i] == s) {
            return 1;
        }
    }
    return 0;
}

int equivalent(int s1, int s2) {
    if (is_final(s1) != is_final(s2)) {
        return 0;
    }
    for (int j = 0; j < n_symbols; j++) {
        if (transition[s1][j] != transition[s2][j]) {
            return 0;
        }
    }
    return 1;
}

void substitute_state(int old_state, int new_state) {
    for (int i = 0; i < n_states; i++) {
        for (int j = 0; j < n_symbols; j++) {
            if (transition[i][j] == old_state) {
                transition[i][j] = new_state;
            }
        }
    }
}

int main() {
    int i, j;

    printf("Enter number of states: ");
    scanf("%d", &n_states);

    printf("Enter number of input symbols: ");
    scanf("%d", &n_symbols);

    printf("Enter transition table ( $\delta(q_i, \text{symbol}) = q_j$ ):\n");
    for (i = 0; i < n_states; i++) {
        for (j = 0; j < n_symbols; j++) {
            printf(" $\delta(q_{%d}, \%c) =$ ", i, 'a' + j);
            scanf("%d", &transition[i][j]);
        }
    }

    printf("Enter number of final states: ");
    scanf("%d", &n_final);

    printf("Enter final states: ");
    for (i = 0; i < n_final; i++) {
        scanf("%d", &final_states[i]);
    }

    for (i = 0; i < n_states; i++) {
```

```
    if (removed[i]) {
        continue;
    }
    for (j = i + 1; j < n_states; j++) {
        if (!removed[j] && equivalent(i, j)) {
            printf("q%d and q%d are equivalent (same group) → Eliminating q%d\n", i,
                j, j);
            substitute_state(j, i);
            removed[j] = 1;
        }
    }
}

printf("\nMinimized DFA Transition Table:\n");

printf("State\t");
for (j = 0; j < n_symbols; j++) {
    printf("%c\t", 'a' + j);
}
printf("\n");

for (i = 0; i < n_states; i++) {
    if (!removed[i]) {
        printf("q%d\t", i);
        for (j = 0; j < n_symbols; j++) {
            printf("q%d\t", transition[i][j]);
        }
        printf("\n");
    }
}

return 0;
}
```